

Privacy-Preserving Federated Learning with Differential Privacy and Adaptive Knowledge Distillation for Dynamic Non-IID Data

Ling Li¹, Lidong Zhu¹ and Weibang Li²

¹ National Key Laboratory of Wireless Communications, University of Electronic Science and Technology of China, Chengdu 611731, China

² School of Computer Science and Artificial Intelligence, Southwest Minzu University, Chengdu 610041, China

Abstract—Federated learning, as a privacy-preserving distributed learning paradigm, faces three major challenges in practical applications: optimization difficulties caused by non-independent and identically distributed (Non-IID) data, distribution shifts introduced by new clients joining in dynamic environments, and privacy threats such as model inversion attacks. Existing methods often address individual challenges at the expense of performance in other aspects, with the conflict between privacy protection and model performance being particularly prominent. This paper proposes a privacy-preserving federated learning framework for dynamic Non-IID data based on differential privacy and adaptive knowledge distillation (DP-ADKD-FL), which addresses multiple challenges through three innovative mechanisms: (1) a sensitivity-based parameter-layered differential privacy mechanism that provides differentiated protection for parameters of varying importance, minimizing the impact of noise on performance; (2) an adaptive knowledge distillation framework that dynamically adjusts knowledge transfer strategies based on client distribution characteristics, effectively responding to distribution changes; and (3) a dynamic privacy budget regulation strategy that optimizes privacy resource allocation according to training stages and the degree of distribution changes. Experiments on real-world datasets demonstrate that DP-ADKD-FL not only significantly outperforms existing methods in privacy protection metrics such as defense against inversion attacks, but also surpasses baseline methods without privacy protection in model performance, especially showing excellent adaptability in dynamic environments.

Keywords—Federated Learning, Differential Privacy, Knowledge Distillation, Non-IID, Privacy Protection

I. INTRODUCTION

As artificial intelligence technologies rapidly develop, data privacy and security protection have become a focal point for both academia and industry. Federated learning [1], as a distributed machine learning paradigm that protects data privacy through the principle of "models move, data stays," enables model training while ensuring user data remains local, and is thus widely considered an effective approach to resolving the conflict between privacy and intelligence requirements. However, federated learning in real-world scenarios faces multiple challenges [2]: on one hand, client data often exhibits non-independent and identically distributed (Non-IID) characteristics, causing inconsistent model optimization trajectories that affect the convergence and generalization capabilities of the global model; on the other hand, federated learning systems typically evolve dynamically, with new clients joining further exacerbating the heterogeneity of data distribution and posing greater challenges to model training. Furthermore, although federated learning itself does not directly share raw data, research indicates that attackers may still be able to reconstruct

sensitive client information through model inversion, gradient leakage, and other attack methods, constituting serious privacy threats.

Existing research has made progress in addressing these challenges [3]. For Non-IID problems, researchers have proposed strategies such as knowledge distillation, representation alignment, and regularization; for privacy protection, differential privacy has been widely applied to federated learning systems, providing strict privacy guarantees for the model update process. However, these methods often lead to performance degradation in other aspects when solving a single problem, especially differential privacy mechanisms which, while enhancing privacy protection, inevitably introduce noise that significantly reduces model performance. More critically, existing methods are mostly designed for static scenarios and struggle to effectively address the challenges brought by client distribution changes in dynamic environments. Therefore, how to protect privacy while addressing dynamic Non-IID environments and achieving an optimal balance among these three factors has become a core issue in the current federated learning field that urgently needs to be resolved.

This paper proposes a privacy-preserving federated learning framework for dynamic Non-IID data based on differential privacy and adaptive knowledge distillation (DP-ADKD-FL), which aims to simultaneously solve the aforementioned multiple challenges. The main innovations of this framework are reflected in the following three aspects:

Sensitivity-based parameter-layered differential privacy mechanism: Unlike traditional methods that apply uniform noise to all model parameters as a coarse-grained protection strategy, this paper categorizes parameters based on their sensitivity to model performance and designs differentiated privacy protection strategies for different layers of parameters, minimizing noise interference on key parameters while maintaining privacy guarantees.

Adaptive knowledge distillation framework: For dynamic Non-IID environments, we propose an adaptive knowledge distillation mechanism based on model uncertainty, which can dynamically adjust knowledge transfer strategies according to client data distribution characteristics and global model states, effectively mitigating distribution shift problems between clients, with special optimization for knowledge fusion processes when new clients join, significantly enhancing model robustness in distribution-changing environments.

Dynamic privacy budget regulation strategy: We propose an adaptive privacy budget allocation method based on training stages and the degree of distribution changes, appropriately increasing the privacy budget in the early stages of model training and during dramatic distribution changes,

while reducing the privacy budget in stable model stages, achieving optimized configuration of privacy resources and obtaining better model performance under the same overall privacy budget constraints.

II. RELATED WORK

A. Federated Learning and Non-IID Challenges

Federated learning is a distributed learning paradigm that allows multiple participants to collaboratively train machine learning models without sharing raw data. McMahan et al. [4] first proposed the Federated Averaging (FedAvg) algorithm, which implements distributed model training through a combination of client local training and server aggregation. However, non-independent and identically distributed (Non-IID) data distribution is widely considered one of the main challenges facing federated learning [5]. Zhao et al. [6] revealed the significant negative impact of Non-IID data on FedAvg performance through empirical studies, attributing it to the inconsistency of model update directions among clients.

To address Non-IID challenges, researchers have proposed various solutions. Li et al. [7] proposed the FedProx algorithm, which limits client model deviation from the global model by introducing a proximity term. Karimireddy et al. [8] designed the SCAFFOLD method, introducing control variables to correct client update bias. Knowledge distillation, as another effective method, has been applied to federated learning environments to mitigate distribution heterogeneity issues. Lin et al. [9] proposed the FedDF framework, using knowledge distillation to integrate knowledge from multiple local models. However, these methods are mainly designed for static environments and have limited adaptability to dynamically changing client distributions. Li et al. [10] recently focused on dynamic federated learning scenarios but did not consider privacy protection issues, whereas this paper simultaneously addresses the triple challenges of dynamic environments, Non-IID data, and privacy protection.

B. Privacy Protection in Federated Learning

Although federated learning avoids directly sharing raw data, research shows that model parameters or gradients may still leak sensitive information. Zhu et al. [11] demonstrated the feasibility of such attacks by reconstructing training samples through gradient inversion. Fredrikson et al. [12] proposed model inversion attacks capable of extracting training data features from machine learning models. To counter these threats, differential privacy has become the primary technology for privacy protection in federated learning.

Differential privacy ensures that the influence of individual data on the final model is limited by adding carefully calibrated random noise to model updates. Abadi et al. [13] proposed the Differentially Private Stochastic Gradient Descent (DP-SGD) algorithm, providing differential privacy guarantees for deep learning models. Geyer et al. [14] introduced differential privacy into federated learning, proposing the DP-FedAvg algorithm. However, these methods typically apply uniform noise to all model parameters, resulting in serious performance losses. Yu et al. [15] attempted to mitigate this problem through adaptive noise scaling but did not consider differences in parameter sensitivity. In contrast, the parameter-layered differential privacy mechanism proposed in this paper considers the

different impacts of parameters on model performance, achieving a more refined privacy-utility balance.

C. Applications of Knowledge Distillation in Federated Learning

Knowledge distillation was initially proposed by Hinton et al. [16] for transferring knowledge from complex models to simple models. In recent years, knowledge distillation has been widely applied to federated learning to mitigate Non-IID problems. Li et al. [17] proposed FedMD, promoting knowledge sharing between clients through mutual distillation on public datasets. Zhu et al. [18] designed the FedGen framework, using generative models to synthesize representative samples for knowledge distillation. However, these methods typically rely on public datasets or generative models, limiting their applicability, and do not address privacy protection issues.

On the other hand, research combining knowledge distillation with privacy protection is relatively limited. Papernot et al. [19] proposed the PATE framework, achieving knowledge transfer under differential privacy protection through teacher ensemble and selective label release. Zhu et al. [20] studied methods to optimize the distillation process under differential privacy constraints. However, these works mainly target centralized scenarios and are difficult to directly apply to dynamic federated environments. The adaptive knowledge distillation framework proposed in this paper not only considers privacy protection constraints but can also dynamically adjust knowledge transfer strategies according to client distribution characteristics and global model states, with special optimization for knowledge fusion processes when new clients join, providing a new solution for dynamic Non-IID federated learning.

III. PRIVACY-PRESERVING FEDERATED LEARNING

A. Privacy-Preserving Federated Learning Framework for Dynamic Non-IID Data

This paper proposes a privacy-preserving federated learning framework for dynamic Non-IID data environments, aimed at solving the triple challenges faced by traditional federated learning: data distribution drift, privacy protection, and performance degradation caused by Non-IID data. The proposed framework includes four key components: a differentially private adaptive knowledge distillation aggregation mechanism, information entropy-based distribution drift monitoring and dynamic privacy budget regulation, heterogeneous encryption and layered transmission mechanism, and cross-node knowledge transfer based on mutual information maximization, as shown in Figure 1.

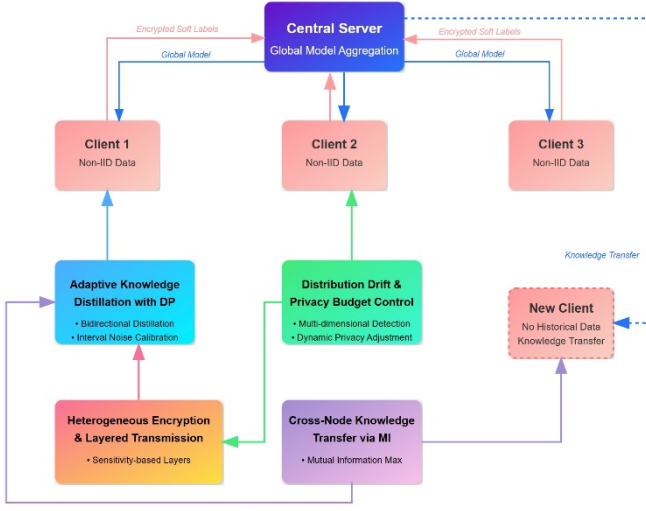


Fig. 1. Privacy-Preserving Federated Learning Framework.

B. Differentially Private Adaptive Knowledge Distillation Aggregation Mechanism

In dynamic Non-IID environments, the main privacy challenge faced by traditional federated learning is that direct transmission of model parameters or gradients is susceptible to model inversion attacks, leading to training data leakage. To address this issue, this paper proposes a differentially private adaptive knowledge distillation mechanism that protects privacy by transmitting differentially private prediction outputs rather than model parameters.

Assume there are K clients, with each client k owning a local dataset $D_k = \{(x_i^k, y_i^k)\}_{i=1}^{n_k}$ and a local model f_{θ_k} . In traditional federated learning, clients directly upload model parameters θ_k or gradients $\nabla \theta_k$. The method proposed in this paper implements secure aggregation through bidirectional knowledge distillation.

Client (Teacher) $\rightarrow$ Server (Student) Distillation:

Client k produces prediction probability outputs for a public dataset D_{pub} or locally generated unlabeled data $D_{unlabeled}^k$:

$$q_k(x) = \text{softmax}(f_{\theta_k}(x) / \tau) \quad (1)$$

where x is the model input data, θ_k represents the model parameters of client k , f_{θ_k} is the local model of client k , and τ is a temperature parameter controlling the smoothness of the probability distribution.

Before uploading prediction probabilities, clients apply a differential privacy mechanism to add noise to the prediction probabilities:

$$\tilde{q}_k(x) = q_k(x) + \mathcal{N}(0, \sigma_k^2 \cdot I) \quad (2)$$

where σ_k is the standard deviation of privacy noise, $\mathcal{N}(0, \sigma_k^2 \cdot I)$ is Gaussian noise with mean 0 and variance σ_k^2 , and I is the identity matrix.

The server aggregates predictions from multiple clients to obtain global teacher knowledge:

$$q_{global}(x) = \sum_{k=1}^K w_k \cdot \tilde{q}_k(x) \quad (3)$$

where w_k is the client weight, K is the total number of clients, and $\tilde{q}_k(x)$ is the noisy prediction probability uploaded by the client.

Server (Teacher) $\rightarrow$ Client (Student) Distillation:

The server trains a global student model f_ϕ , with the optimization objective:

$$\mathcal{L}_{global} = \sum_{x \in D_{pub}} D_{KL}(q_{global}(x) || p_\phi(x)) \quad (4)$$

where $p_\phi(x) = \text{softmax}(f_\phi(x) / \tau)$ is the prediction probability of the global model, D_{pub} is the public dataset, D_{KL} is the Kullback-Leibler divergence, and ϕ is the global model parameters.

The server distributes the trained global model parameters ϕ to each client, and client k combines global knowledge with local supervised learning, with its local optimization objective as follows:

$$\mathcal{L}_k = (1 - \alpha) \cdot \mathcal{L}_{CE}(f_{\theta_k}, D_k) + \alpha \cdot \mathcal{L}_{KD}(f_{\theta_k}, f_\phi, D_k \cup D_{unlabeled}^k) \quad (5)$$

where $\mathcal{L}_{CE}$ is the cross-entropy loss, $\mathcal{L}_{KD}$ is the knowledge distillation loss, α is a balancing factor, and $D_{unlabeled}^k$ is the unlabeled dataset of client k .

The client model optimization objective is as follows:

$$\theta_k^* = \arg \min_{\theta_k} \mathcal{L}_k \quad (6)$$

This paper proposes an interval propagation noise calibration method, which further reduces the required noise amount for the differential privacy mechanism by strictly limiting the range interval of prediction probabilities. Specifically, an interval mapping is applied to $q_k(x)$ before adding noise:

$$\hat{q}_k(x) = \text{clip}(q_k(x), a, b) \quad (7)$$

The global sensitivity after interval mapping becomes:

$$\Delta \hat{q} = (b - a) \cdot \sqrt{2} \quad (8)$$

This significantly reduces the required noise amount and improves model accuracy.

C. Information Entropy-Based Distribution Drift Monitoring and Dynamic Privacy Budget Regulation

This paper proposes a multi-dimensional distribution drift detection method that comprehensively considers changes in feature distribution, label distribution, and feature-label joint distribution.

Define the category distribution of client k at time t as $P_k^t = [p_1^t, p_2^t, \dots, p_C^t]$, where p_i^t represents the proportion of category i . The Jensen-Shannon divergence is used to measure changes in category distribution:

$$\Delta_{label,k}^t = D_{JS}(P_k^t \| P_k^{t-1}) \quad (9)$$

where P_k^t represents the category distribution of client k at time t , P_k^{t-1} is the category distribution of client k at time $t-1$.

The Maximum Mean Discrepancy (MMD) is used to measure changes in feature distribution:

$$\Delta_{feature,k}^t = \text{MMD}(X_k^t, X_k^{t-1}) \quad (10)$$

Finally, conditional distribution changes are used to capture changes in feature-label relationships:

$$\Delta_{cond,k}^t = \frac{1}{C} \sum_{c=1}^C D_{JS}(P(X|Y=c)^t \| P(X|Y=c)^{t-1}) \quad (11)$$

This paper proposes an information theory-based dynamic privacy budget regulation mechanism that associates distribution drift with privacy budget. The privacy budget $\mathcal{E}_k^t$ of client k at time t is defined as follows:

$$\mathcal{E}_k^t = \mathcal{E}_{base} \cdot (1 + \beta \cdot g(\Delta_k^t)) \quad (12)$$

where $\mathcal{E}_{base}$ is the base privacy budget, β is a regulation coefficient, and $g(\Delta_k^t)$ is a non-linear mapping function.

Based on the privacy budget $\mathcal{E}_k^t$, the noise standard deviation of client k at time t is dynamically adjusted as follows:

$$\sigma_k^t = \frac{\widetilde{\Delta q}_k \cdot \sqrt{2 \ln(1.25 / \delta)}}{\mathcal{E}_k^t} \quad (13)$$

D. Heterogeneous Encryption and Layered Transmission Mechanism

This paper proposes a sensitivity-based parameter layering method that divides model parameters into three levels—high sensitivity, medium sensitivity, and low sensitivity—and adopts different privacy protection strategies for each.

A parameter sensitivity function $S(\theta_{k,i})$ is defined to measure the importance of parameter $\theta_{k,i}$ to data privacy, specifically as follows:

$$S(\theta_{k,i}) = \alpha \cdot \|\nabla_{\theta_{k,i}} \mathcal{L}\|_2 + \beta \cdot I(\theta_{k,i}; D_k) + \gamma \cdot V(\theta_{k,i}) \quad (14)$$

where $\|\nabla_{\theta_{k,i}} \mathcal{L}\|_2$ is the norm of the parameter gradient, $I(\theta_{k,i}; D_k)$ is the mutual information between the parameter and the dataset, $V(\theta_{k,i})$ is the variance of the parameter over multiple rounds of training, and α, β, γ are weight coefficients.

According to sensitivity, parameters are divided into the following three layers:

$$\Theta_k^{\text{high}} = \{\theta_{k,i} \mid S(\theta_{k,i}) > \tau_{\text{high}}\} \quad (15)$$

$$\Theta_k^{\text{mid}} = \{\theta_{k,i} \mid \tau_{\text{low}} < S(\theta_{k,i}) \leq \tau_{\text{high}}\} \quad (16)$$

$$\Theta_k^{\text{low}} = \{\theta_{k,i} \mid S(\theta_{k,i}) \leq \tau_{\text{low}}\} \quad (17)$$

Different encryption strategies are adopted for parameters with different sensitivity levels.

For high-sensitivity parameters Θ_k^{high} , Fully Homomorphic Encryption (FHE) is adopted:

$$\text{Enc}_{FHE}(\Theta_k^{\text{high}}) = \text{FHE.Encrypt}(pk, \Theta_k^{\text{high}}) \quad (18)$$

For medium-sensitivity parameters Θ_k^{mid} , Functional Encryption (FE) is adopted:

$$\text{Enc}_{FE}(\Theta_k^{\text{mid}}) = \text{FE.Encrypt}(mpk, \Theta_k^{\text{mid}}) \quad (19)$$

For low-sensitivity parameters Θ_k^{low} , Local Differential Privacy (LDP) is used for encryption:

$$\Theta_k^{\text{low,priv}} = \Theta_k^{\text{low}} + \text{Lap}(\Delta \Theta_k^{\text{low}} / \mathcal{E}_k^t) \quad (20)$$

For high-sensitivity parameters, the properties of fully homomorphic encryption are utilized to aggregate directly in ciphertext space:

$$\text{Agg}(\Theta^{\text{high}}) = \text{FHE.Eval}\left(\sum_{k=1}^K w_k \cdot \text{Enc}_{FHE}(\Theta_k^{\text{high}})\right) \quad (21)$$

Aggregation of medium-sensitivity parameters is implemented through functional encryption to achieve weighted summation:

$$\text{Agg}(\Theta^{\text{mid}}) = \text{FE.Decrypt}(sk_f, \text{Enc}_{FE}(\Theta_k^{\text{mid}})_{k=1}^K) \quad (22)$$

For low-sensitivity parameters, the parameters with added noise are directly aggregated:

$$\text{Agg}(\Theta^{\text{low}}) = \sum_{k=1}^K w_k \cdot \Theta_k^{\text{low,priv}} \quad (23)$$

Finally, the federated learning server obtains the complete global model parameters:

$$\Theta_{\text{global}} = [\text{Agg}(\Theta^{\text{high}}), \text{Agg}(\Theta^{\text{mid}}), \text{Agg}(\Theta^{\text{low}})] \quad (24)$$

E. Cross-Node Knowledge Transfer Based on Mutual Information Maximization

This paper proposes a knowledge distillation framework based on mutual information maximization, aimed at maximizing the mutual information between teacher and student models while minimizing privacy leakage risk.

The mutual information between the teacher model f_T and the student model f_S is defined as follows:

$$I(f_T; f_S) = H(f_T) - H(f_T | f_S) \quad (25)$$

To maximize effective knowledge transfer while minimizing privacy leakage, the optimization objective is designed as follows:

$$\max_{f_S} I(f_T; f_S) - \lambda \cdot I(D_{private}; f_S) \quad (26)$$

For newly joined nodes, this paper designs an adaptive transfer learning mechanism to achieve secure and efficient knowledge transfer. Denote the new node as j with a local small sample dataset D_j . First, obtain the global student model f_ϕ from the server, then fine-tune it by combining local supervised learning and knowledge distillation.

To further enhance the adaptability of new nodes, this paper introduces public domain data to assist in training. Denote the public domain dataset as D_{pub} , and the new node enhances adaptation through the following steps:

Pseudo-Label Generation: Use the global model f_ϕ to generate pseudo-labels $\hat{y}_i$ for public data:

$$\hat{y}_i = \arg \max_c f_\phi(x_i), \forall x_i \in D_{pub} \quad (27)$$

Confidence Filtering: Select high-confidence samples to build an auxiliary training set, specifically represented as:

$$D_{pub}^{high} = \{(x_i, \hat{y}_i) \mid \max_c f_\phi(x_i)_c > \tau_{conf}, x_i \in D_{pub}\} \quad (28)$$

Domain Adaptation Training: Combine local data and high-confidence public data for training, with the domain adaptation training objective for the new node represented as:

$$\mathcal{L}_j^{adapt} = \mathcal{L}_j + \lambda \cdot \mathcal{L}_{CE}(f_{\theta_j}, D_{pub}^{high}) \quad (29)$$

IV. EXPERIMENTAL EVALUATION

A. Experimental Setup

Experimental Environment: The simulation experiments were developed in PyCharm 17.0.10 environment, using Python 3.9 as the programming language and PyTorch 2.3 as the deep learning framework. For the simulation experiment settings, 10 clients were configured, and for Non-IID scenario construction, imbalanced label distribution, feature distribution shifts, and dynamic distribution change simulations were set up for the simulation experiments.

Datasets: The experiments used the MNIST dataset [21] and the EuroSAT dataset [22]. The MNIST dataset contains handwritten digit images from 0 to 9 and is a standard benchmark dataset widely used to evaluate the performance of various algorithms. The EuroSAT dataset is a remote sensing image dataset based on satellite imagery, including 10 different types of land use.

B. Evaluation Metrics

For model performance metrics, this paper evaluates based on indicators such as Accuracy and Average Loss.

The accuracy formula is represented as follows:

$$Acc = \frac{TP + TN}{TP + TN + FP + FN} \quad (30)$$

where TP (True Positive) is the number of true positives, TN (True Negative) is the number of true negatives, FP (False Positive) is the number of false positives, and FN (False Negative) is the number of false negatives.

The Average Loss formula is as follows:

$$L_{avg} = \frac{1}{N} \sum_{i=1}^N L(y_i, \hat{y}_i) \quad (31)$$

For privacy protection metrics, evaluation is based on indicators such as Differential Privacy Budget.

The differential privacy budget formula is as follows:

$$\epsilon_{total} = \log \left(1 + \sum_{k=1}^K w_k^2 (e^{\epsilon_{kt}} - 1) \right) \quad (32)$$

where w_k is the weight of client k , ϵ_{kt} is the privacy budget of client k at time t .

C. Experimental Results and Evaluation

The baselines selected in this paper are the traditional federated averaging method FedAvg [4], the federated averaging method with differential privacy DP-FedAvg [14], the knowledge distillation federated learning method FedDF [9], and the distribution-adaptive federated learning method SCAFFOLD [8], with our method represented as DP-ADKD-FL.

Model Performance Metric Evaluation:

Experiments were conducted on both the MNIST and EuroSAT datasets, training for 20 rounds with new clients introduced in the 5th round.

Figures 2 and 3 show comparisons of accuracy and average loss on the MNIST dataset among FedAvg, DP-FedAvg, FedDF, SCAFFOLD, and our proposed method.

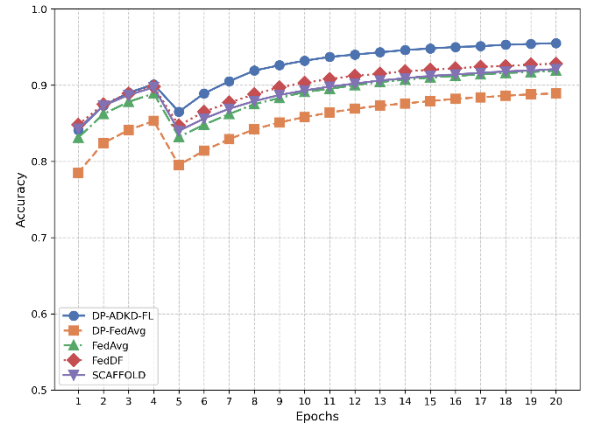


Fig. 2. Accuracy Comparison on MNIST Dataset.

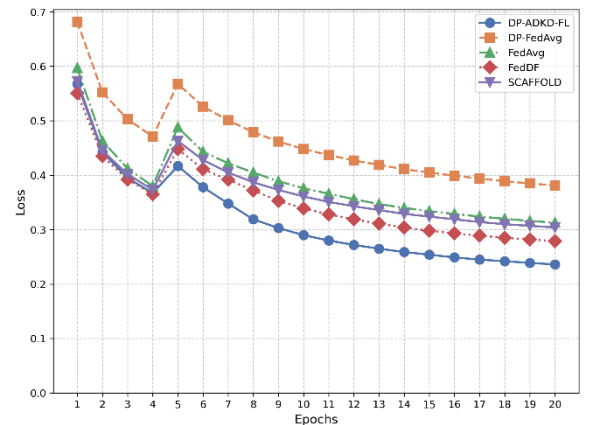


Fig. 3. Average Loss Comparison on MNIST Dataset.

As shown in Figure 2, throughout the 20 communication rounds, DP-ADKD-FL significantly outperforms the other four methods in terms of accuracy. Notably, while performance was comparable across all methods during the initial training phase (rounds 1-4), DP-ADKD-FL demonstrated exceptional adaptability and recovery speed after new clients were introduced in round 5. At round 5, all methods experienced performance degradation, reflecting the impact of distribution shifts introduced by new clients.

Figure 3 illustrates the loss function curves of the five methods on the MNIST dataset, reflecting the optimization behavior and convergence characteristics of each method in a dynamic Non-IID environment. During the initial training phase, all methods experienced rapid loss reduction. After the introduction of new clients in round 5, all methods underwent a significant rebound in loss values; comparatively, DP-ADKD-FL exhibited the smallest increase in loss, demonstrating its exceptional stability when facing distribution shifts.

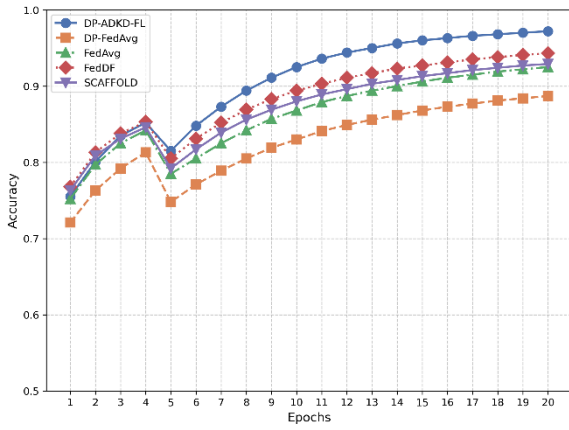


Fig. 4. Accuracy Comparison on EuroSAT Dataset.

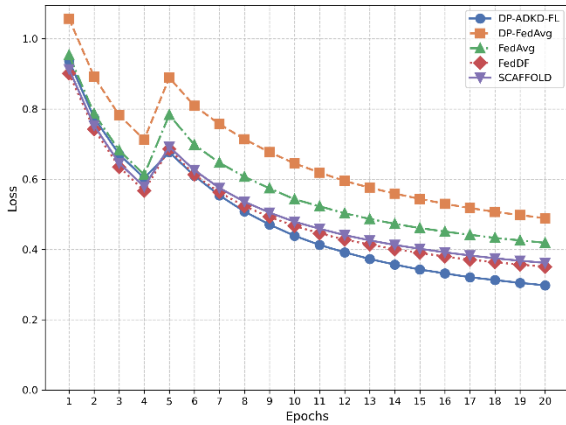


Fig. 5. Average Loss Comparison on EuroSAT Dataset.

As observed in Figure 4, the five methods exhibit distinct performance differences on the EuroSAT dataset. DP-ADKD-FL demonstrates the most prominent performance throughout the training process, especially in handling distribution shifts. Specifically, performance disparities were relatively small during the initial training phase (rounds 1-4), but rapidly expanded after new clients were introduced in round 5. At round 5, all methods experienced performance degradation, but DP-ADKD-FL's accuracy decreased noticeably less than other methods. This result indicates that DP-ADKD-FL possesses significant distributional robustness and can more

effectively adapt to data distribution changes introduced by new clients. Even more notably, during the recovery phase following new client introduction, DP-ADKD-FL demonstrated exceptionally rapid adaptation capabilities.

Figure 5 displays the loss function curves of the five methods on the EuroSAT dataset. From the overall trend, all methods show generally decreasing loss trajectories across both datasets. In general, DP-ADKD-FL performs best in terms of the loss function, and its advantage expands as data complexity increases.

Privacy Protection Effectiveness Evaluation:

Figure 6 demonstrates the performance of various methods on the inversion MSE metric based on the EuroSAT dataset. This metric measures the difference between samples obtained by attackers through model inversion and the original samples, with higher values indicating lower quality of inverted samples and better privacy protection. DP-ADKD-FL performs exceptionally well on this metric, suggesting that even if an attack succeeds, the information obtained from the samples would be extremely limited. This demonstrates the superiority of DP-ADKD-FL in disrupting sample reconstruction.

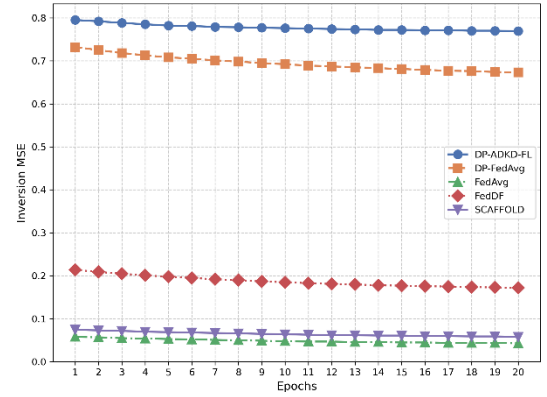


Fig. 6. Performance on Inversion MSE Metric Based on EuroSAT Dataset.

Figure 7 compares the model inversion attack success rates across different methods based on the EuroSAT dataset. It clearly shows that the DP-ADKD-FL method exhibits excellent privacy protection capabilities. Throughout the entire training process, DP-ADKD-FL consistently maintains the lowest inversion attack success rate. This indicates that DP-ADKD-FL's parameter-layered privacy protection strategy can more effectively resist model inversion attacks.

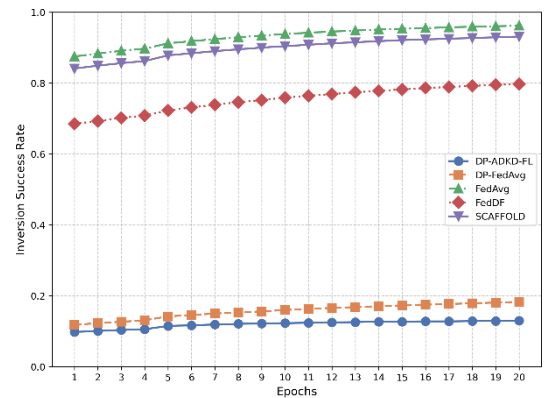


Fig. 7. Comparison of Inversion Attack Success Rates.

V. CONCLUSION

This paper proposes a privacy-preserving federated learning framework for dynamic Non-IID data based on differential privacy and adaptive knowledge distillation (DP-ADKD-FL). Through three innovations—a sensitivity-based parameter-layered privacy protection mechanism, an adaptive knowledge distillation framework, and a dynamic privacy budget regulation strategy—it not only outperforms existing methods in model inversion defense effectiveness but also breaks the traditional trade-off between privacy protection and model performance, achieving synchronous improvement in both. This paper validates the effectiveness of the proposed method on real-world datasets. Future research will explore privacy protection strategies in adversarial environments to address more complex attack patterns.

Acknowledgement: This research was funded by the National Natural Science Foundation of China under Grant 62371098.

REFERENCES

- [1] Zhang C, Xie Y, Bai H, et al. A survey on federated learning. *Knowledge-Based Systems*, 2021, 216: 106775.
- [2] Wen J, Zhang Z, Lan Y, et al. A survey on federated learning: challenges and applications. *International Journal of Machine Learning and Cybernetics*, 2023, 14(2): 513-535.
- [3] Zhang K, Song X, Zhang C, et al. Challenges and future directions of secure federated learning: a survey. *Frontiers of computer science*, 2022, 16: 1-8.
- [4] McMahan B, Moore E, Ramage D, et al. Communication-efficient learning of deep networks from decentralized data. *Artificial intelligence and statistics*. PMLR, 2017: 1273-1282.
- [5] Kairouz P, McMahan H B, Avent B, et al. Advances and open problems in federated learning. *Foundations and trends® in machine learning*, 2021, 14(1-2): 1-210.
- [6] Zhao Y, Li M, Lai L, et al. Federated learning with non-iid data. *arXiv preprint arXiv:1806.00582*, 2018.
- [7] Li T, Sahu A K, Zaheer M, et al. Federated optimization in heterogeneous networks. *Proceedings of Machine learning and systems*, 2020, 2: 429-450.
- [8] Karimireddy S P, Kale S, Mohri M, et al. Scaffold: Stochastic controlled averaging for federated learning. *International conference on machine learning*. PMLR, 2020: 5132-5143.
- [9] Lin T, Kong L, Stich S U, et al. Ensemble distillation for robust model fusion in federated learning. *Advances in neural information processing systems*, 2020, 33: 2351-2363.
- [10] Li Q, Diao Y, Chen Q, et al. Federated learning on non-iid data silos: An experimental study. *2022 IEEE 38th international conference on data engineering (ICDE)*. IEEE, 2022: 965-978.
- [11] Zhu L, Liu Z, Han S. Deep leakage from gradients. *Advances in neural information processing systems*, 2019, 32.
- [12] Fredrikson M, Jha S, Ristenpart T. Model inversion attacks that exploit confidence information and basic countermeasures. *Proceedings of the 22nd ACM SIGSAC conference on computer and communications security*. 2015: 1322-1333.
- [13] Abadi M, Chu A, Goodfellow I, et al. Deep learning with differential privacy. *Proceedings of the 2016 ACM SIGSAC conference on computer and communications security*. 2016: 308-318.
- [14] Geyer R C, Klein T, Nabi M. Differentially private federated learning: A client level perspective. *arXiv preprint arXiv:1712.07557*, 2017.
- [15] Yu D, Zhang H, Chen W, et al. Do not let privacy overbill utility: Gradient embedding perturbation for private learning. *arXiv preprint arXiv:2102.12677*, 2021.
- [16] Hinton G, Vinyals O, Dean J. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.
- [17] Li D, Wang J. Fedmd: Heterogeneous federated learning via model distillation. *arXiv preprint arXiv:1910.03581*, 2019.
- [18] Zhu Z, Hong J, Zhou J. Data-free knowledge distillation for heterogeneous federated learning. *International conference on machine learning*. PMLR, 2021: 12878-12889.
- [19] Papernot N, Abadi M, Erlingsson U, et al. Semi-supervised knowledge transfer for deep learning from private training data. *arXiv preprint arXiv:1610.05755*, 2016.
- [20] Zhu L, Liu Z, Han S. Deep leakage from gradients. *Advances in neural information processing systems*, 2019, 32.
- [21] LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11), 2278-2324.
- [22] Li L, Zhu L, Li W. Cloud-Edge-End Collaborative Federated Learning: Enhancing Model Accuracy and Privacy in Non-IID Environments. *Sensors*, 2024, 24(24): 8028.